다음 함수에 대한 진리표를 보아라

a. $F = XZ + X'Y' + X'Y'Z$

b. $G = Y'Z + (X+Z)\cdot(X+Y)$

c. $H = W'X + WY + ZXY + WXY + WY'Z'$

8. 속성 1부터 10까지 이용하여 다음식을 최소 곱의 식으로 줄여라.
각 단계를 보여라. (최소항의 수와 리터럴의 수가 괄호 안에 주어졌다).

a. $xy + x'yz + xyz$ (2항 4리터럴)

b. $xyz' + yz + y'z$ (2항 4리터럴)

c. $xz + x'y + xyz + xyz'$ (2항 3리터럴)

d. $ab + a'b' + abc + ab'c$ (3항 6리터럴)

e. $xyz + xy'z + xz + x'yz$ (2항 4리터럴)

f. $xy'z + x'y'z + xyz + xy + x'z'$ (3항 6리터럴)

g. $xyz + xyz' + xy'z + xy'z + xyz'$ (2항 5리터럴)

h. $ab'c + a'b'c + abc' + ab'c' + a'bc + abc$ (2항 3리터럴)

10. 다음함수를 구현하기 위한 시스템의 회로도를 AND, OR, NOT 게이트를 사용하여 나타내라. 변수들은 보편화되지 않은 것만 사용 가능하다고 가정한다. 대수식은 조작하지 않는다

a. $g = PR + PQ + Q'$

b. $f = abc + a(b+c)$

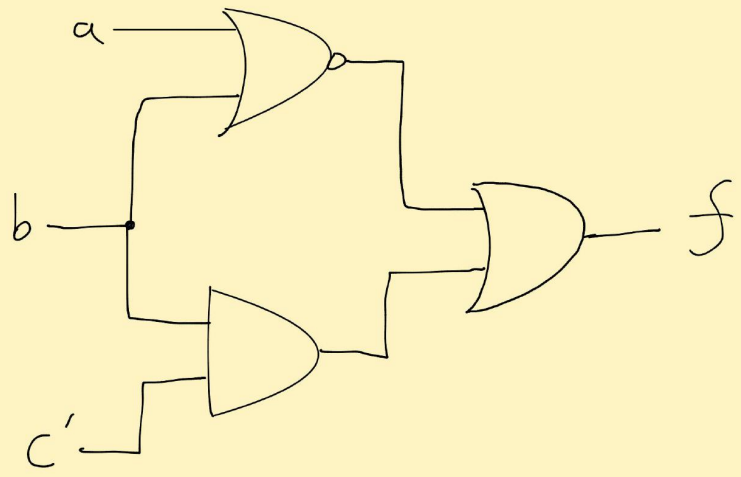
c. $h = XY(W + Z') + (V + WZ)(X + Y)'$

18. 다음 각 회로에 대해

i) 대수식을 구해라

ii) 그것을 곱의 합으로 나타내어라.

1)



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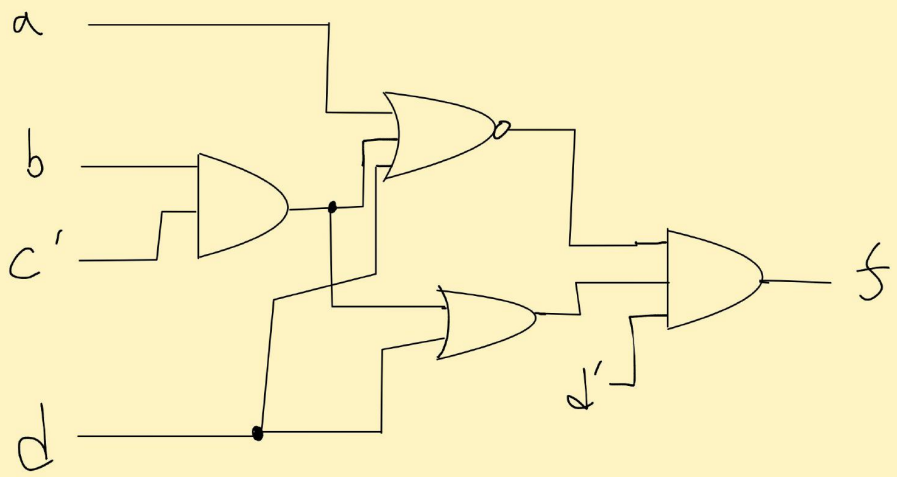

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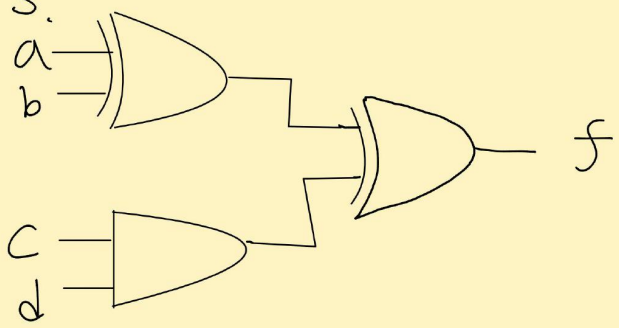

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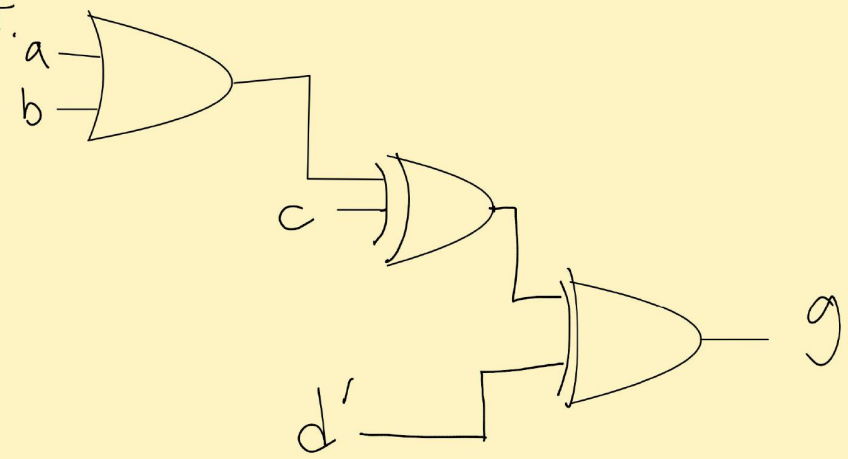

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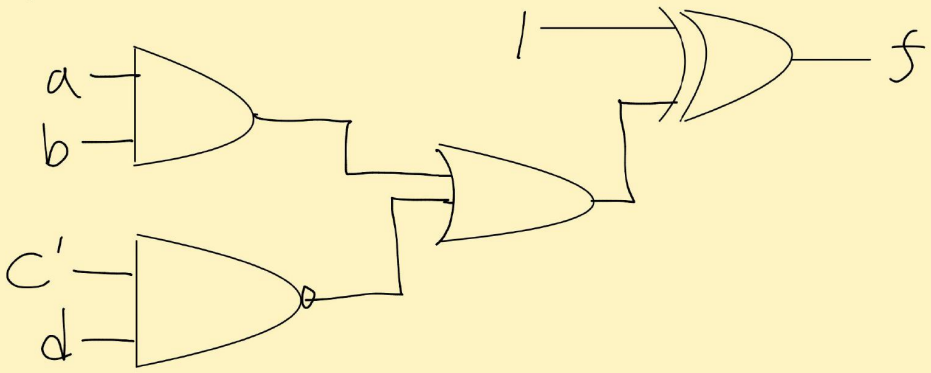

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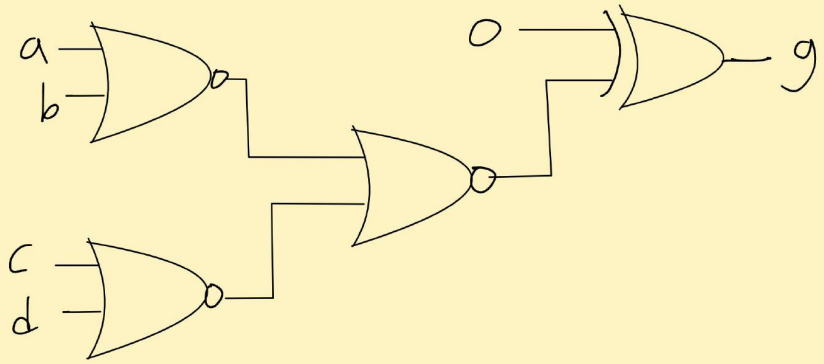


2:

X	Y	Z	X'	Y'	Z'	XZ	X'Y	X'Y'Z	F
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	0	1	1
0	1	0	1	0	1	0	1	0	1
0	1	1	1	0	0	0	1	0	1
1	0	0	0	1	1	0	0	0	0
1	0	1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	0	0	1

b.

X	Y	Z	X'	Y'	Z	YZ	(X+Z)	(X+Y)	(X+Z)(X+Y)	G
0	0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	1	1	0	0	1
0	1	0	1	0	1	0	0	1	0	0
0	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1
1	0	1	0	1	0	1	1	1	1	1
1	1	0	0	0	1	0	1	1	1	1
1	1	1	0	0	0	0	1	1	1	1

C.

W	X	Y	Z	W'X	WY	ZXY	WXY	WX'Z'	H
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	1	0	0	0	0	1
0	1	0	1	1	0	0	0	0	1
0	1	1	0	1	0	0	0	0	1
0	1	1	1	1	0	1	0	0	1
1	0	0	0	0	0	0	0	1	1
1	0	0	1	0	0	0	0	1	1
1	0	1	0	0	1	0	0	0	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0
1	1	1	0	0	1	0	1	0	1
1	1	1	1	0	1	1	1	0	1

8.

a. $xy + x'yz + xyz$

$= yz(x'+x) + xy$

$= yz + xy$

b. $xyz' + yz + y'z$

$= xyz' + z(y+y')$

$= xyz' + z$

c. $xz + x'y + xyz + xyz'$

$= xz + x'y + xy(z+z')$

$= xz + y(x'+x)$

$= xz + y$

d. $ab + a'b' + abc + ab'c$

$= ab + a'b' + ac(b+b')$

$= ab + a'b' + ac$

e. $xyz + xy'z + xz + x'yz$

$= (xyz + xy'z) + xz + x'yz$

$= xz(y+y') + x + x'yz$

$= xz + z(x + x'y)$

$= xz + z(x+y)$

$= xz + xz + zy$

$= xz + zy$

f. $xy'z + x'y'z + xyz + xy + x'z'$

$= (xy'z + x'y'z) + (xyz + xy) + x'z'$

$= y'z(x+x') + xy(z+1) + x'z'$

$= y'z + xy + x'z'$

g. $xyz + xyz' + xy'z + xy'z + xyz'$

$= (xyz + xy'z) + (xyz' + xyz') + xy'z$

$= xz(y+y') + xyz' + xy'z$

$= xz(1+y') + xyz' + xy'z$

$= xz(1+y') + xyz'$

$= xz + xyz'$

h. $ab'c + a'b'c + abc' + ab'c' + a'bc + abc$

$= b'c(a+a') + ac'(b+b') + bc(a+a')$

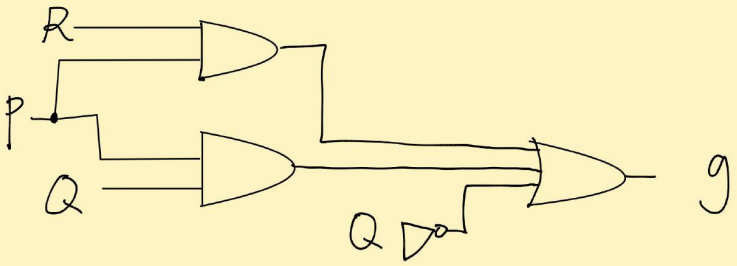
$= b'c + ac' + bc$

$= c(b'+b) + ac'$

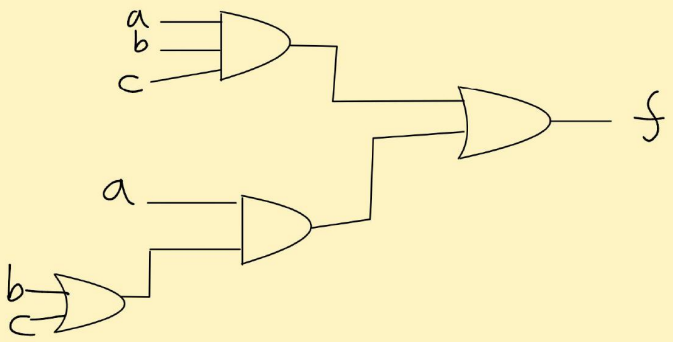
$= c + ac'$

10.

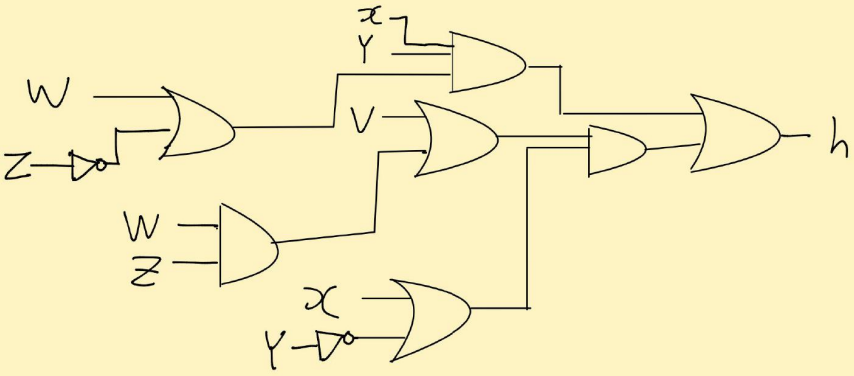
a. $PR + PQ + Q'$



b. $f = abc + a(b+c)$



c. $h = xy(w+z') + (v+wz)(x+y')$



18.

1)

i) $f = (a+b)' + bc'$

$= a'b' + bc'$

ii) $a'b' + bc'$

2)

i) $f = (a + b\cdot c' + d)' + (bc' + d) + d'$

$= a'\cdot(b'+c)\cdot d' + (bc'+d) + d$

$= a'\cdot b'\cdot d' + a'c\cdot d' + bc' + d + d$

ii) $a'b'd' + a'c\cdot d' + bc' + d$

$= a'd'(b'+c) + bc' + d$

3.

i) $f = \underbrace{a \oplus b}_{p} \oplus \underbrace{(c\cdot d)}_{z}$ $\quad p = a'b + ab'$

ii) $\Rightarrow p \oplus z = p'z + pz'$ $\quad p' = (a \oplus b)' = a'b' + ab$

$= (a'b' + ab)(c\cdot d) + (a'b + ab')\cdot(c' + d')$

$= a'b'cd + abcd + a'bc' + ab'c' + a'bd' + ab'd'$

4. i) $g = \underbrace{(a+b)}_{z} \oplus \underbrace{c \oplus d'}_{p}$

$P = c'd' + cd$

ii) $\Rightarrow z \oplus p = z'p + zp'$

$z' = (a+b)' = a'b'$

$= a'b'(c'd'+cd) + (a+b)(c'd+cd')$

$p' = c'd + cd'$

$= ab'c'd' + ab'cd + ac'd + acd' + bc'd + bcd'$

5. i) $f = (a \cdot b + (c' \cdot d)') \oplus 1$

$= (a'+b') \cdot (c' \cdot d) \oplus 1$

$= (a'c'd + b'c'd) \oplus 1$

ii) $(a'c'd + b'c'd)'$

$= (a + \underline{c+d'}) \cdot (b + \underline{c+d'})$

$= ab + a(c+d') + b(c+d') + c+d'$

$= ab + ac + ad' + bc + bd' + c + d'$

$= ab + c + d'$

6. i) $g = ((a+b)' + (c+d)')' \oplus 0$

$= (a+b) \cdot (c+d) \oplus 0$

$= ac + ad + bc + bd \oplus 0$

ii) $ac + ad + bc + bd$